

Codelingo

Complete Frontend Architecture, UI Component Directory & System Specification

EXECUTIVE SUMMARY

Codelingo is a Duolingo-inspired, gamified programming education web application. It combines adaptive AI pedagogical coaching (featuring **Louis the Cardinal**), real-time CodeMirror interactive coding sandboxes, psychometric learner mode branching (Swell integration), and a Duolingo-style staggered learning path. This document provides an exhaustive, production-grade technical inventory of every page, component, progression algorithm, API integration, and design token operating on the **dev** branch.

GIT BRANCH & STATUS

dev (Tracking origin/dev)

LATEST COMMIT

ca3abc4 (Path layout & icons fix)

FRAMEWORK & RUNTIME

React 19.2 • Vite 8.3 • TypeScript 6.0

STYLING & UI PRIMITIVES

Tailwind CSS v4 • Framer Motion v13 •
CodeMirror 6

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Core Frontend Philosophy: The Codelingo frontend is designed around zero-friction micro-learning. Every exercise is chunked into 2-5 minute interactions, immediately verified via CodeMirror and regex evaluators, with continuous positive reinforcement delivered through animated streak fire, XP gains, and Louis the Cardinal reactions.

1 Executive Architecture & Tech Stack

The Codelingo client is a Single Page Application (SPA) built on Vite and React 19. It uses modern declarative state patterns, hardware-accelerated animations, and dark/light system theme sync.

Core UI Foundation

React 19.2

High-concurrency React 19 utilizing client-side routing with react-router-dom v7. Component state is clean, modular, and leverages custom hooks for theme and responsive layouts.

Server State & Cache

TanStack Query v5

Asynchronous data fetching, automatic background revalidation, query invalidation upon code evaluation, and seamless fallback to local mock data if the API backend is unreachable.

Interactive Code Editor

CodeMirror 6

Powered by @uiw/react-codemirror with language extensions for Python and JavaScript/TypeScript. Features dark syntax themes, gutter line numbers, and active line glow.

Micro-Interactions & FX

Framer Motion & Confetti

Fluid physics-based spring animations for Louis the Cardinal mascot, expandable progress accordions, pulsating nodes, and multi-particle dual-cannon canvas fireworks upon lesson completion.

Technology Inventory

PACKAGE / TECHNOLOGY	VERSION	PRIMARY ROLE IN FRONTEND
react & react-dom	^19.2.8	Application component hierarchy, hooks, and DOM mounting
react-router-dom	^7.18.3	Client routes: /learn, /lesson/:lang/:id, /complete

PACKAGE / TECHNOLOGY	VERSION	PRIMARY ROLE IN FRONTEND
@tanstack/react-query	^5.102.8	Server state caching, mutation triggers, and optimistic UI refetching
tailwindcss & @tailwindcss/vite	^4.3.3	Next-generation utility CSS engine with native dark-mode selectors
@uiw/react-codemirror	^4.25.11	Embedded multi-language IDE code editor widget
framer-motion	^13.2.0	Declarative layout animations, mascot emotions, modal transitions
lucide-react	^1.45.0	High-clarity SVG icon library for lesson topics, tracks, and UI controls
canvas-confetti	^1.9.4	Celebration confetti particle explosions on correct solutions

2 Application Routing & Screen Hierarchy

The application operates on three primary views connected via `react-router-dom`. Each page is designed for high accessibility, responsive mobile-to-desktop scaling, and zero layout shift.

1. Learn Page (Dashboard / Track Navigator)

Route: `/learn`

The central hub where users select their active language, view their learning path, inspect their gamification statistics, and launch lessons.

- **Top Course Banner:** Displays active programming language label, total percentage, and lesson progress counter.
- **Swell Personality Badge:** Summarizes user's active cognitive trait (e.g. ANALYTICAL) and fit score (e.g. 88%).
- **Louis Coach Greeting:** Warm mascot welcome with personalized message matching user profile.
- **Duolingo-Style Learning Path:** Staggered alternating 2-column lesson nodes with central connected progress line.
- **Side Sticky Progress Column:** Houses the interactive `ProgressRecord` with dynamic red-to-green meter.

2. Lesson Page (Interactive Learning & Sandbox)

Route: `/lesson/:language/:lessonId`

The core pedagogical sandbox where the user learns concepts and solves programming exercises.

- **Sticky Header:** Animated `LessonProgressBar` (0% → 50% → 100%) and `ThemeToggle`.
- **Lesson Topic Header:** Shows language badge, lesson number out of 10, and topic icon (e.g., `GitBranch` for Conditions).
- **Demo Personality Switcher:** Quick switcher bar enabling judges to test Analytical, Practical, and Visual modes in real time.
- **Adaptive Exercise Engine:** Renders `DEEP_EXPLANATION` (`ConceptCard` + `CodeMirror`), `PRACTICE_FIRST` (Fill-in-the-Blank), or `VISUAL_GUIDED` (Flowchart + Guided Fill).
- **Bottom Action Sheet (FeedbackCard):** Fixed bottom tray providing Check, Try Again, and Continue triggers with diagnostic error feedback.

3. Complete Page (Milestone Celebration)

Route: `/complete`

High-dopamine payoff screen triggered upon completing an exercise.

- **Dual-Cannon Fireworks:** Automatic celebration confetti bursts from both left and right screen margins.
- **Giant Celebrating Mascot:** Louis the Cardinal jumping with floating party poppers (🎉) and sparkles (✨).
- **Stat Cards:** Animated Flame Daily Streak (+1 day) and Lightning XP Reward (+10 XP) cards.
- **Continue CTA:** Primary button navigating the learner back to the updated learning path.

3 Complete UI Component Directory

The frontend codebase is organized into atomic, highly reusable components in `src/components/`:

CATEGORY	COMPONENT	FILE LOCATION	PRIMARY FUNCTIONALITY
Layout & Navigation	AppShell	components/layout/AppShell.tsx	Root layout wrapper managing header, content max-width, and padding
	TopNavigation	components/navigation/TopNavigation.tsx	Sticky brand header with logo, course selector, badges, and account CTA
	CourseSelector	components/navigation/CourseSelector.tsx	Dropdown menu to switch between 8 programming languages across 4 categories
	ThemeToggle	components/navigation/ThemeToggle.tsx	Button toggling dark and light mode with sun/moon icon animations
Learning Path	LearningPath	components/path/LearningPath.tsx	Zigzag 2-column progression tree with central connector progress bar
	LessonNode	components/path/LessonNode.tsx	Circular button node with completed, current (pulse ring), and locked states
Lesson Sandbox	CodeExercise	components/lesson/CodeExercise.tsx	CodeMirror IDE editor with syntax highlighting, gutter, and starter code
	FillBlankExercise	components/lesson/FillBlankExercise.tsx	Interactive code snippet with inline blank text input and error shake
	ConceptCard	components/lesson/ConceptCard.tsx	Card rendering technical concept explanations, bullets, and visual flowcharts
	FeedbackCard	components/lesson/FeedbackCard.tsx	Sticky bottom tray displaying check button, validation state, and feedback text
	LessonProgressBar	components/lesson/LessonProgressBar.tsx	Top horizontal progress bar tracking lesson completion percentage
Gamification & AI	LouisCoach	components/louis/LouisCoach.tsx	Mascot widget with 4 mood states, bouncing animations, and speech bubbles
	ProgressRecord	components/gamification/ProgressRecord.tsx	Side card with dynamic red-to-green progress bar, milestones, and all tracks
	StreakBadge	components/gamification/StreakBadge.tsx	Pill badge showing active flame streak count with pop animation
	XpBadge	components/gamification/XpBadge.tsx	Pill badge showing accumulated XP points with lightning bolt
	PersonalityBadge	components/personality/PersonalityBadge.tsx	Pill displaying learner psychometric trait, learning mode, and score
	DemoPersonalitySwitch	components/demo/DemoPersonalitySwitch.tsx	Floating quick switcher for live demonstrations of adaptive modes

CATEGORY	COMPONENT	FILE LOCATION	PRIMARY FUNCTIONALITY
Auth & Profile	CreateAccountModal	components/auth/CreateAccountModal.tsx	Registration dialog with name, email, password, and instant account creation

4 Louis Mascot & Adaptive Psychometrics

Codelingo's signature feature is **Louis the Cardinal**, an interactive AI pedagogical coach paired with Swell psychometric cognitive analysis. Louis provides contextual, emotionally intelligent feedback at every stage.

Louis Coach Mascot States

LouisMood

- idle:** Subtle vertical hovering motion (2.6s loop) while learner browses.
- thinking:** Head-tilt rotation with floating lightbulb overlay (💡?) during deep explanations.
- encouraging:** Gentle breathing pulse with motivational tips before submission.
- celebrating:** Energetic bouncing jump (y: -16px, rotate: ±6°), with party popper (🎉) and sparkle (✨) particles.

Dynamic Speech Bubble

Framer Motion

Features an authentic pointing triangular arrow anchored directly to Louis's beak. Whenever Louis's message updates, the bubble performs a smooth fade-and-slide micro-transition (opacity: 0 → 1, y: 6 → 0) preventing jarring text jumps.

Swell Psychometric Learner Modes

Depending on the user's cognitive profile, the lesson dynamically restructures its presentation mode:

TRAIT	MODE KEY	PEDAGOGICAL STRATEGY	RENDERED COMPONENT FLOW
ANALYTICAL	DEEP_EXPLANATION	Theory first, rigorous syntax rules, step-by-step logic breakdown	ConceptCard (in-depth) → CodeExercise (Full CodeMirror)
PRACTICAL	PRACTICE_FIRST	Immediate hands-on challenge, minimal reading, trial-and-error	Immediate FillBlankExercise → Instant execution feedback
VISUAL	VISUAL_GUIDED	Flowcharts, step-by-step visual logic diagrams, color cues	ConceptCard (Flowchart steps) → Guided FillBlankExercise

5 Progression Engine & Red-to-Green Dynamic System

Codelingo features a custom-engineered **Dynamic Red-to-Green Color Progression Engine** located in `ProgressRecord.tsx`. As the user completes lessons, the UI's progress bars, badges, and mastery indicators smoothly transition from crimson novice red to vibrant emerald mastery green.

Progress Tier & Color Transition Scale

Interpolated Gradient

0-20% Novice (Crimson) 20-40% Apprentice (Orange) 40-60% Intermediate (Amber) 60-80% Proficient (Lime) 80-100% Mastered (Emerald)

Gradient Engine Formula

`ProgressRecord.tsx`

```
const getProgressColor = (percent: number) => {
  if (percent >= 80) return "#10b981"; // Emerald
  if (percent >= 60) return "#84cc16"; // Lime
  if (percent >= 40) return "#f59e0b"; // Amber
  if (percent >= 20) return "#f97316"; // Orange
  return "#ef4444"; // Crimson Red
};
```

Earned Milestones System

4 Core Badges

- **First Steps:** Unlocked upon completing 1st lesson (CheckCircle2 icon).
- **Flame Keeper:** Maintained a 4+ day streak (Flame icon).
- **Centurion XP:** Earned over 100 XP points (Zap lightning icon).
- **Polyglot Coder:** Explored 3 or more tracks (Trophy icon).

All Tracks Overview Grid

The bottom of `ProgressRecord` contains a compact 2x4 card grid tracking real-time completion across all 8 languages, complete with language icons and independent mini red-to-green progress bars.

6 Curriculum Engine & Supported Languages

Codelingo supports 8 premier programming languages grouped into 4 distinct developer career tracks:

TRACK	BADGE	CATEGORY	ACCENT COLOR	CURRICULUM FOCUS
C#	C#	Enterprise	#9333EA	Enterprise backend & game development with C# .NET
JavaScript	JS	Web & Frontend	#F59E0B	Web frontend, dynamic scripting & full-stack Node.js
TypeScript	TS	Web & Frontend	#2563EB	Type-safe modern web, large-scale apps & APIs
Python	Py	Popular	#3B82F6	Data science, AI scripting & readable backend logic
Rust	Rs	Systems & Cloud	#EA580C	Ultra-fast systems programming, memory safety & WASM
Go (Golang)	Go	Systems & Cloud	#06B6D4	Microservices, cloud infrastructure & high concurrency
C++	C++	Systems & Cloud	#1D4ED8	Game engines, graphics, low latency & performance
Java		Enterprise	#DC2626	Enterprise architecture, Spring Boot & Android apps

10-Stage Progressive Lesson Templates

STAGE #	LESSON ID	TITLE	SYNTAX TAG	TOPIC & CONCEPT COVERED
1	hello	1. Hello World	>_	Console I/O, syntax basics, entry points
2	variables	2. Variables & Types	var	Constants, primitive types, memory allocation
3	conditions	3. Conditions	if	Boolean logic gates, if/else branch control
4	functions	4. Functions	fn()	Parameters, return types, functional scope
5	loops	5. Loops & Iteration	for	While, for-in, repetition and break guards
6	arrays	6. Arrays & Slices	[]	Collections, zero-indexing, dynamic slices
7	oop	7. Structs & Classes	class	Object modeling, encapsulation, constructors
8	async	8. Async & Concurrency	async	Tasks, promises, coroutines, thread safety
9	errors	9. Error Handling	try	Try/catch, Result types, panic recovery
10	generics	10. Generics & Traits	<T>	Type parameters, protocols, interfaces, reuse

7 State Management, API Client & Mock Layer

Codelingo features a resilient hybrid networking layer. If the backend server is running on `http://localhost:3000`, real HTTP REST requests are dispatched. If the backend is unreachable or offline, the client automatically falls back to an extensive, deterministic in-memory mock engine (`mockData.ts`).

React Query Cache Keys

@tanstack/react-query

- ["dashboard", userId]: Fetches user stats, active tracks, and streaks.
- ["personality", userId]: Retrieves Swell cognitive scores and modes.
- ["adaptiveLesson", lang, lessonId, trait]: Fetches customized lesson content.

Theme Context & Dark Mode

ThemeContext.tsx

Maintains theme state across sessions. Persists to `localStorage` under `codelingo-theme` ("dark" | "light") and automatically toggles the dark class on `document.documentElement` for Tailwind v4 CSS variables.

REST Endpoints Matrix

METHOD	ENDPOINT	FRONTEND HOOK / FUNCTION	DESCRIPTION
GET	/api/dashboard?userId=...	fetchDashboard(userId)	Fetches user profile, streaks, and course progress for all 8 tracks
GET	/api/personality?userId=...	fetchPersonality(userId)	Fetches Swell cognitive scores, primary trait, and learning mode
GET	/api/lessons/:lang/:id	fetchAdaptiveLesson(lang, id, uid)	Fetches tailored lesson payload with CodeMirror starter code & prompts
POST	/api/evaluate	evaluateExercise(payload)	Evaluates submitted user code or fill-in-the-blank answers
POST	/api/demo/personality	setDemoPersonality(userId, trait)	Sets mock personality trait for instant live demonstration switching

8 Source Code File Matrix

Comprehensive inventory of all 42 source files residing under `codingo-web/src/`:

RELATIVE PATH	LINES	CATEGORY	KEY EXPORTED SYMBOLS & RESPONSIBILITIES
<code>main.tsx</code>	21	Entry	Root render with <code>QueryClientProvider</code> and <code>ThemeProvider</code>
<code>App.tsx</code>	44	Router	Browser router routing to <code>/learn</code> , <code>/lesson/:lang/:id</code> , <code>/complete</code>
<code>App.css</code> & <code>index.css</code>	130	Styles	Global resets, Tailwind imports, scrollbars, and focus rings
<code>pages/LearnPage.tsx</code>	121	Page	<code>LearnPage</code> component: Path, Louis coach, and sticky <code>ProgressRecord</code>
<code>pages/LessonPage.tsx</code>	369	Page	<code>LessonPage</code> component: Mode switching, <code>CodeMirror</code> , and <code>FeedbackCard</code>
<code>pages/CompletePage.tsx</code>	146	Page	<code>CompletePage</code> component: Celebration fireworks, streak & XP cards
<code>components/layout/AppShell.tsx</code>	54	Layout	<code>AppShell</code> : Standardized wrapper with sticky header
<code>components/navigation/TopNavigation.tsx</code>	104	Nav	<code>TopNavigation</code> : Header with course selector, streak, and XP
<code>components/navigation/CourseSelector.tsx</code>	118	Nav	<code>CourseSelector</code> : Dropdown with categorized language tracks
<code>components/navigation/ThemeToggle.tsx</code>	48	Nav	<code>ThemeToggle</code> : Dark/light mode button toggle
<code>components/path/LearningPath.tsx</code>	129	Path	<code>LearningPath</code> : 2-column zigzag tree with middle progress line
<code>components/path/LessonNode.tsx</code>	112	Path	<code>LessonNode</code> : Circular button with completed/current/locked states
<code>components/lesson/CodeExercise.tsx</code>	91	Lesson	<code>CodeExercise</code> : <code>CodeMirror</code> dark code editor
<code>components/lesson/FillBlankExercise.tsx</code>	88	Lesson	<code>FillBlankExercise</code> : Interactive code snippet with blank input
<code>components/lesson/ConceptCard.tsx</code>	115	Lesson	<code>ConceptCard</code> : Markdown explanation and visual flowcharts
<code>components/lesson/FeedbackCard.tsx</code>	138	Lesson	<code>FeedbackCard</code> : Bottom action sheet with validation banners
<code>components/lesson/LessonProgressBar.tsx</code>	42	Lesson	<code>LessonProgressBar</code> : Top animated horizontal progress meter
<code>components/louis/LouisCoach.tsx</code>	123	Mascot	<code>LouisCoach</code> : Cardinal mascot with 4 mood states & speech bubble
<code>components/gamification/ProgressRecord.tsx</code>	422	Gamification	<code>ProgressRecord</code> : Red-to-green dynamic progress card
<code>components/gamification/StreakBadge.tsx</code>	45	Gamification	<code>StreakBadge</code> : Flame daily streak indicator
<code>components/gamification/XpBadge.tsx</code>	45	Gamification	<code>XpBadge</code> : Lightning bolt XP points counter
<code>components/personality/PersonalityBadge.tsx</code>	58	Gamification	<code>PersonalityBadge</code> : Swell cognitive trait pill
<code>components/demo/DemoPersonalitySwitch.tsx</code>	62	Utility	<code>DemoPersonalitySwitch</code> : Live switcher for demo judges
<code>components/auth/CreateAccountModal.tsx</code>	132	Auth	<code>CreateAccountModal</code> : User registration dialog
<code>api/client.ts</code>	45	API	Base HTTP fetch client with error normalization

RELATIVE PATH	LINES	CATEGORY	KEY EXPORTED SYMBOLS & RESPONSIBILITIES
api/dashboard.ts	32	API	Dashboard query function with mock fallback
api/lessons.ts	48	API	Adaptive lesson query function with mock fallback
api/evaluate.ts	52	API	Code and blank solution evaluation mutation
api/personality.ts	36	API	Personality query function with mock fallback
api/demo.ts	28	API	Demo personality update mutation
api/mockData.ts	480	Mock	Deterministic fallback data for all 8 tracks and 10 lessons
context/ThemeContext.tsx	64	Context	Theme provider syncing dark mode to document element
context/useTheme.ts	18	Context	useTheme hook helper
lib/constants.ts	152	Lib	SUPPORTED_LANGUAGES and LESSON_TEMPLATES definitions
lib/icons.tsx	130	Lib	LessonTopicIcon and LanguageTrackIcon mappings
types/api.ts	88	Types	TypeScript interfaces for API requests and payloads
types/lesson.ts	15	Types	LouisMood, SubmitStatus, and PathLessonNode types

Summary of Frontend Assets: In addition to code files, the frontend bundles essential static graphic assets in public/ and assets/ including louis-transparent.png (official mascot), logo.png (Codelingo brand emblem), and language track badges.